

Computational Interaction Science Citation Map

This map links canonical concepts to likely supporting bibliography entries. Mappings are corpus-derived and require human confirmation before publication.

Domain commitments

A set of commitments about admissible entities, categories, relations, identity conditions, modalities, and constraints in a domain. In computational work, these commitments may be approximated through ontologies, schemas, contracts, types, or model definitions, but none of those artifacts is automatically a philosophical ontology.

Origin: Ontology, Semantics, Representation, and Interpretation

Candidate sources: parasuraman2000; wc2023; wiener1948cybernetics; wittgenstein1953; simon1955; simon1955a; grice1957; simon1962

System state

The physical, computational, social, or modeled condition of a declared system at a particular time.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: shannon1948a; wiener1948cybernetics; hyman1953; fitts1954; bertalanffy1968; bertalanffy1968general; vonbertalanffy1968; norman1986

Semantic assignment

A rule-governed assignment of content, denotation, truth conditions, inferential consequences, or transition behavior to expressions, states, signs, or operations within a specified system and domain.

Origin: Ontology, Semantics, Representation, and Interpretation

Candidate sources: wc2023; tarski1944; scott1971; yarkoni2017; morris1938; skinner1938; rosenblueth1943; shannon1948a

Semantic state

The content-bearing assignments and relations currently licensed by a specified semantics for a system or discourse state.

Origin: Ontology, Semantics, Representation, and Interpretation

Candidate sources: tarski1944; wc2023; shannon1948a; barhillel1953; fitts1954; bertalanffy1968; bertalanffy1968general; vonbertalanffy1968

Information

A typed measure or characterization of distinctions, correlations, descriptions, or content relative to a specified source, code, semantic relation, probability model, or use.

Origin: Ontology, Semantics, Representation, and Interpretation

Candidate sources: barhillel1953; floridi2004; kolchinsky2018; judd1908; tarski1944; hick1952; hyman1953; fitts1954

Representation

A materially or formally realized vehicle that provides selective access to a declared target through a mapping or correspondence and an interpretive practice.

A representation may preserve, transform, omit, or introduce structure.

Origin: Representation, Invariance, and Projection

Candidate sources: wc2023; gentner1983; tarski1944; kuhn1970; palmer1978; salmon1984; lave1988; ericsson1993

Projection

A declared transformation from a target description, system state, or semantic assignment into an accessible representational form.

A Projection Contract specifies:

- target and authority;
- mapping or construction rule;
- interpretation assumptions;
- preserved properties;
- transformed properties;
- omitted properties;
- introduced properties;
- available operations;
- observer assumptions;
- intended use;
- context and scope;
- uncertainty, tolerance, and failure conditions.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: tarski1944; cousot1977; stoffregen2003; shimojima2015; wc2023; shannon1948a; barhillel1953; fitts1954

Scoped invariance

A declared property remains invariant across projections when independently specified comparison rules show that it is preserved for a stated target, observer class, context, metric, and tolerance.

Invariance must be typed. Relevant types include:

- ontological;
- semantic;
- informational;
- structural;
- topological;

computational;
inferential;
task;
behavioral;
interactional.

Origin: Canonical glossary

Candidate sources: wc2023; tarski1944; yarkoni2017; wiener1948cybernetics; barhillel1953; simon1955; simon1955a; searle1980

Interpretation

The application, recovery, construction, negotiation, or revision of semantic relations by an interpreting system under a context, competence, history, and purpose.

Origin: Ontology, Semantics, Representation, and Interpretation

Candidate sources: cousot1977; rosenblueth1943; tarski1944; shannon1948a; barhillel1953; fitts1954; bertalanffy1968; bertalanffy1968general

Interaction participant

Any bounded person, collective, organism, computational process, artificial system, institution, or hybrid arrangement occupying a consequential role within an interaction.

A participant may observe, interpret, act, constrain, authorize, mediate, represent, store, or transform state. Participation does not imply autonomy, intelligence, consciousness, or learning.

Origin: Canonical glossary

Candidate sources: lave1991; beer1995; sutton1999; yarkoni2017; wc2023; fikes1971; rothenberg1989; cohen1990

Agent

An acting interaction participant that selects or produces actions through a policy, rule, control process, intention, or other action-selection organization.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: wc2023; suchman1987plans2; suchman1987plans3; beer1995; goodwin2000; dourish2001; dourish2001where; fipa2002

Environment

The declared physical, computational, social, symbolic, or mixed system within which action and state transition occur.

Origin: Behavior

Candidate sources: goodwin2000; stoffregen2003; anderson2007; yarkoni2017; wc2023; shannon1948a; wiener1948cybernetics; fitts1954

Affordance

An action-relevant relation between an agent's effectivities or capabilities and environmental structure that makes a class of action possible. In the Gibsonian sense, perception is not required for the affordance to exist.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: wc2023; bunke1997; yarkoni2017; judd1908; kuhn1970; gibson1979; gibson1979the; gibson1979the2

Opportunity

A typed and indexed analytical profile coordinating action-availability relations under declared system, agent, projection, task, authority, state, and temporal conditions.

Opportunity is not a new primitive. It coordinates established relations such as feasibility, capability, permission, enablement, reachability, exposure, signaling, belief, expected consequence, reversibility, and uncertainty.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: wc2023; yarkoni2017; tarski1944; shannon1948a; fitts1954; bertalanffy1968; bertalanffy1968general; vonbertalanffy1968

Opportunity profile

For an agent, action, state, projection, and time, an opportunity profile records the relevant typed relations: domain-valid;

effectively capable;

permitted;

enabled;

reachable outcomes;

exposed;

signaled;

believed available;

expected consequences;

selected;

executed;

successful;

reversible or recoverable;

uncertainty and provenance.

Selected, executed, and successful are trace predicates rather than availability sets.

Origin: Representation, Invariance, and Projection

Candidate sources: wc2023; yarkoni2017; tarski1944; hyman1953; levenshtein1966; vanglabbeek1990; vanglabbeek1990the; taylor1994

Capability

The effective capacity or resources required for an agent-system coupling to execute an action or bring about an outcome under declared conditions.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: fitts1954; wc2023; shannon1948a; bertalanffy1968; bertalanffy1968general; vonbertalanffy1968; norman1986; norman1986cognitive

Permission

Authorization under a policy, role, rule, authority, or institutional condition for a subject to perform an operation.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: ericsson1993; sandhu1996; ferraiolo2003

Enablement

The condition in which an action's current-state preconditions are satisfied.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: thelen1994; kelso1995; goodwin2000; dourish2001; dourish2001where; vanderaalst2016; vanderaalst2016process; yarkoni2017

Reachability

The existence of at least one admissible transition sequence from a declared initial state to a target state or outcome within specified constraints and horizon.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: thorndike1901; goodwin2000; jastrzembski2007

Exposure

Provision by the current projection of an executable access path to an operation.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: tarski1944; sandhu1996; ferraiolo2003; yarkoni2017; wc2023

Signaling

Perceivable information indicating an operation, condition, consequence, state, or recovery path.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: hick1952; barhillel1953; hyman1953; fitts1954; borko1968; dretske1981; tuftes1983; guarino1998

Discoverability

The empirically observed probability and process by which an agent identifies an action or relation under declared task, projection, knowledge, modality, context, and time conditions.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: wc2023; vanderaalst2016; vanderaalst2016process; yarkoni2017; judd1908; tarski1944; hyman1953; dretske1981

Believed action

An action represented by an agent as available, with an associated confidence and expected consequence.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: wc2023; yarkoni2017; thelen1994; beer1995; goodwin2000; dourish2001; dourish2001where; fipa2002

Action

A declared intervention, command, operation, or behavior capable of participating in a state transition.

Origin: Opportunity, Affordance, and Action Availability

Candidate sources: yarkoni2017; wc2023; skinner1938; rosenblueth1943; wiener1948cybernetics; levenshtein1966; kendon1990; thelen1994

State transition

A change from one declared system state to another under a transition relation, process, or event.

Origin: Interaction

Candidate sources: thorndike1901; judd1908; shannon1948a; fitts1954; bertalanffy1968; bertalanffy1968general; vonbertalanffy1968; norman1986

Observation

A recorded or accessible distinction about state, transition, representation, or outcome available through a specified measurement or observation process.

Origin: Formal Models of Computational Interaction

Candidate sources: yarkoni2017; wc2023; tarski1944; stevens1946; stevens1946on; thorndike1949; suppes1963; krantz1971

Situated behavior

A bounded, temporally extended trajectory of observable or instrumentally recoverable changes, outputs, or regulated persistence attributed to a system in relation to its environment under a declared observation and attribution model.

Origin: Canonical glossary

Candidate sources: stoffregen2003; judd1908; skinner1938; rosenblueth1943; shannon1948a; wiener1948cybernetics; fitts1954; simon1955

Behavioral evidence

A provenance-preserving chain connecting direct observations, system-derived features, latent interpretations, alternative explanations, uncertainty, and falsifying evidence to a bounded research claim.

Origin: Behavior

Candidate sources: wc2023; yarkoni2017; skinner1938; rosenblueth1943; tarski1944; shannon1948a; wiener1948cybernetics; fitts1954

Behavioral equivalence

Equivalence of declared action or outcome distributions for a specified population, task, context, metric, and tolerance.

Origin: Representation, Invariance, and Projection

Candidate sources: abate2017; wc2023; simon1955; simon1955a; searle1980; stichweh1992; thelen1994; kuutti1996

Interactional equivalence

Equivalence of the available operation-transition-observation structure under specified agent, task, projection, authority, state, context, and tolerance conditions.

Origin: INTERACTION AS A SCIENTIFIC EXPLANANDUM

Candidate sources: abate2017; wc2023; tarski1944; kuhn1970; gentner1983; salmon1984; stichweh1992; beer1995

Interaction

A family of bounded coupling phenomena in which distinguishable systems or participants become mutually consequential through perturbation, constraint, uptake, response, and recurrence.

Origin: Interaction

Candidate sources: beer1995; parasuraman2000; hornbk2017what; oulasvirta2022; bell1973; milner1980; park1981; card1983

Interaction episode

A bounded interval in which two or more distinguishable systems are coupled such that perturbations originating in, or mediated by, one system alter the possible or actual state trajectory of another, and the resulting configuration constrains a later trajectory of an originating or co-participating system within a declared boundary and timescale.

Reciprocal closure may be direct, indirect, asymmetric, delayed, probabilistic, or environmentally mediated.

Origin: Interaction

Candidate sources: beer1995; wegner1997; goodwin2000; parasuraman2000; hornbk2017what; oulasvirta2022; thorndike1901; shannon1948a

Coupling

A relation or pathway through which a participant or shared medium can alter another participant's transition structure, reachable states, probabilities, timing, or constraints.

Origin: Interaction

Candidate sources: judd1908; codd1970; kuhn1970; gentner1983; salmon1984; clark1991; bunke1997; maturana2002

Perturbation

A difference propagated through a coupling relation that may change state, transition conditions, probabilities, timing, or reachable outcomes.

Origin: Interaction

Candidate sources: judd1908; bunke1997; maturana2002; sen2016

Uptake

Incorporation of a perturbation into a participant's state, transition process, control process, or interpreted situation. Uptake need not be semantic or conscious.

Origin: Interaction

Candidate sources: tarski1944; wiener1948cybernetics2; wiener1948cybernetics3; wiener1948cybernetics4; barhillel1953; kalman1960; ramadge1987; turvey1992

Recurrence

Return influence through which a response constrains an originating participant, a co-participant, or the shared coupling configuration.

Origin: Interaction

Candidate sources: thorndike1901; codd1970; clark1991; maturana2002

Interaction contract

A declaration of interaction identity, participants and roles, boundary, timescale, state, coupling, preconditions, events, transitions, protocol, invariants, evidence, failure, recovery, and completion conditions.

Origin: Interaction

Candidate sources: fipa2002; davidson1980; park1981; card1983; card1983the; card1983the2; card1983the3; gieryn1983

Interaction morphology

The measurable structural properties of a representation or interaction surface, such as hierarchy, topology, density, alignment, rhythm, symmetry, enclosure, continuity, direction, visual weight, temporal organization, and modality.

Origin: Canonical glossary

Candidate sources: wc2023; yarkoni2017; tarski1944; gower1971; sacks1974; palmer1978; gibson1979; gibson1979the

Interaction grammar

A typed vocabulary and composition structure for semantic actions and transitions independent of any one input modality or rendering implementation.

Origin: CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?

Candidate sources: suchman1987plans2; suchman1987plans3; thorndike1901; tarski1944; barhillel1953; kuhn1970; davidson1980; park1981

Interaction history

The accumulated experience, training, conventions, outcomes, failures, practices, and prior exposure that may change an agent's later beliefs, policies, and interpretation.

Origin: INTERACTION AS A SCIENTIFIC EXPLANANDUM

Candidate sources: wc2023; beer1995; fipa2002; ferrando2019; judd1908; geertz1973; cousot1977; cousot1977abstract

Trace

A temporally ordered or partially ordered record of observable or attributed events under declared boundaries, granularity, and observation rules.

Origin: Interaction Similarity

Candidate sources: kaelbling1998; kaelbling1998planning; wc2023; simon1955; simon1955a; davidson1980; searle1980; park1981

Measurement contract

A declaration of the measurand, observation process, operationalization, scale, uncertainty, validity evidence, population, task, projection, comparison conditions, and permitted interpretation.

Origin: Measurement

Candidate sources: wc2023; taylor1994; jcgm2008; yarkoni2017; tarski1944; stevens1946; stevens1946on; thorndike1949

Interaction similarity

A comparison of interaction tokens, types, structures, protocols, opportunities, traces, behaviors, policies, strategies, outcomes, causes, or measurements under an explicit frame, correspondence mapping, admissible transformations, metrics, tolerances, uncertainty, and validity domain.

Origin: INTERACTION AS A SCIENTIFIC EXPLANANDUM

Candidate sources: yarkoni2017; wc2023; park1981; wooldridge1995; parasuraman2000; ferrando2019; cronbach1955; gower1971

Token identity

Historical identity of one bounded interaction episode under declared continuity, participant, boundary, and timescale criteria.

Origin: Interaction

Candidate sources: thorndike1901; park1981; card1983; card1983the; card1983the2; card1983the3; gieryn1983; hoare1985communicating

Type identity

Membership in the same abstraction-defined interaction class under explicit classification rules.

Origin: Interaction Similarity

Candidate sources: park1981; card1983; card1983the; card1983the2; card1983the3; rasmussen1983; hoare1985communicating; newell1985

Equivalence

Exact preservation under a declared Comparison Contract and observable set. Equivalence is always typed.

Origin: Interaction Similarity

Candidate sources: kaelbling1998; kaelbling1998planning; abate2017

Similarity

Graded proximity under declared dimensions, representations, alignment rules, metrics, weights, tolerances, and uncertainty.

Origin: Interaction Similarity

Candidate sources: wc2023; gower1971; tversky1977; park1981; rasmussen1983; taylor1994; zhang1994; zhang1994representations

Incomparability

Failure to establish a defensible common comparison space because boundaries, targets, mappings, scales, observation processes, uncertainty, or validity domains cannot be aligned without unsupported assumptions.

Origin: Comparative Interaction

Candidate sources: stevens1946; stevens1946on; cronbach1955; hoare1985; hoare1985communicating; ramadge1987; taylor1994; messick1995

Similarity contract

A declaration of comparison target, frame, representations, correspondence mapping, invariants, variants, observables, admissible and prohibited transformations, typed metrics or predicates, tolerances, uncertainty, validity domain, and final typed profile.

Origin: Interaction Similarity

Candidate sources: wc2023; cronbach1955; gower1971; tversky1977; park1981; gentner1983; taylor1994; zhang1994

Strategy

An inferred, temporally extended organization of conditional action selection, information use, resource allocation, and contingency response that tends to preserve or improve one or more criteria across a declared class of interaction states.

Origin: Interaction Strategies

Candidate sources: wc2023; wegner1997; goodwin2000; dourish2001; dourish2001where; hartson2003; spence2007; fidel2012

Strategy contract

A declaration of strategy-bearing system, domain, timescale, criteria, state and observation model, state abstraction, opportunity assumptions, action repertoire, selection policy, information policy, commitments, contingency and recovery rules, switching and stopping conditions, resources, authority, identity invariants, rival strategies, uncertainty, and validity domain.

Origin: Interaction Strategies

Candidate sources: yarkoni2017; wc2023; fitts1954; shannon1948a; thorndike1949; hick1952; barhillel1953; hyman1953

Policy

A conditional relation from represented states, observations, or beliefs to action distributions.

Origin: Interaction Strategies

Candidate sources: judd1908; norman1983; thelen1994; bunke1997; goodwin2000; dourish2001; dourish2001where; vandraalst2016

Fundamental

A representation-independent computational substrate for declaring domain commitments, evolving system state, assigning operational semantics, and producing governed projections whose preservation, transformation, omission, introduced structure, operations, snapshots, and replay can be inspected.

Origin: CAN INTERACTION BECOME A PRIMARY SCIENTIFIC OBJECT?

Candidate sources: wc2023; tarski1944; palmer1978; yarkoni2017; shannon1948a; wiener1948cybernetics; fitts1954; bertalanffy1968

Computational Interaction Science

A research program that studies interaction as an explanatory target by separating and comparing declared domain commitments, state, semantics, projection, representation, interpretation, opportunity, coupling, behavior, measurement, similarity, strategy, and evidence.

CIS must justify itself through comparative explanatory gain, empirical validity, reproducible methods, and error-sensitive predictions. It does not replace HCI, psychology, anthropology, cognitive science, design, computer science, or related disciplines.

Origin: Computational Interaction Science

Candidate sources: yarkoni2017; wc2023; opensciencecollaboration2015; tarski1944; park1981; newell1976; card1983; card1983the